ANDROID PROGRAMMING PRACTICAL 14

Module 2 · Media APIs and Telephone APIs

Aim

Implement Media APIs and Telephone APIs.

Project name: MediaTelephoneDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a UI with separate controls for a media feature and a telephone action.

2. Add a small audio resource under res/raw if demonstrating MediaPlayer playback.

3. Initialize MediaPlayer safely, start playback on button click, and release it in lifecycle cleanup.

4. For telephone functionality, use an ACTION_DIAL intent with a tel: URI so the user confirms the call in the dialer.

5. Avoid placing calls directly unless explicitly required and appropriately permissioned.

6. Test media playback and dialer launch on an emulator or physical device with suitable support.

7. Handle missing media resources, unavailable dialer apps, and lifecycle cleanup.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The media demo works and the telephone action opens the dialer with a prepared number.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.